Day 8 Task: Capture Analysis

Warm-up - get your bearings

Question	Your answer
How many packets are in the capture? (bottom bar)	67
Statistics → Protocol Hierarchy: which protocols appear?	Ethernet, IPv4, TCP, UDP, HTTP, DNS,
Statistics → Endpoints: how many IPv4 addresses are involved?	6

Finding 1 - The plaintext login

Question	Your answer
Which client IP sent the login?	10.0.0.15
Which server IP received it?	10.0.0.20
What is the username?	admin
What is the password? (it's URL-encoded)	Tekl00my%40123
Why was this readable, and what one change would hide it?	It was readable because HTTP sends data in plain text without encryption. Switching to HTTPS (TLS encryption) would hide it.

Finding 2 - The port scan

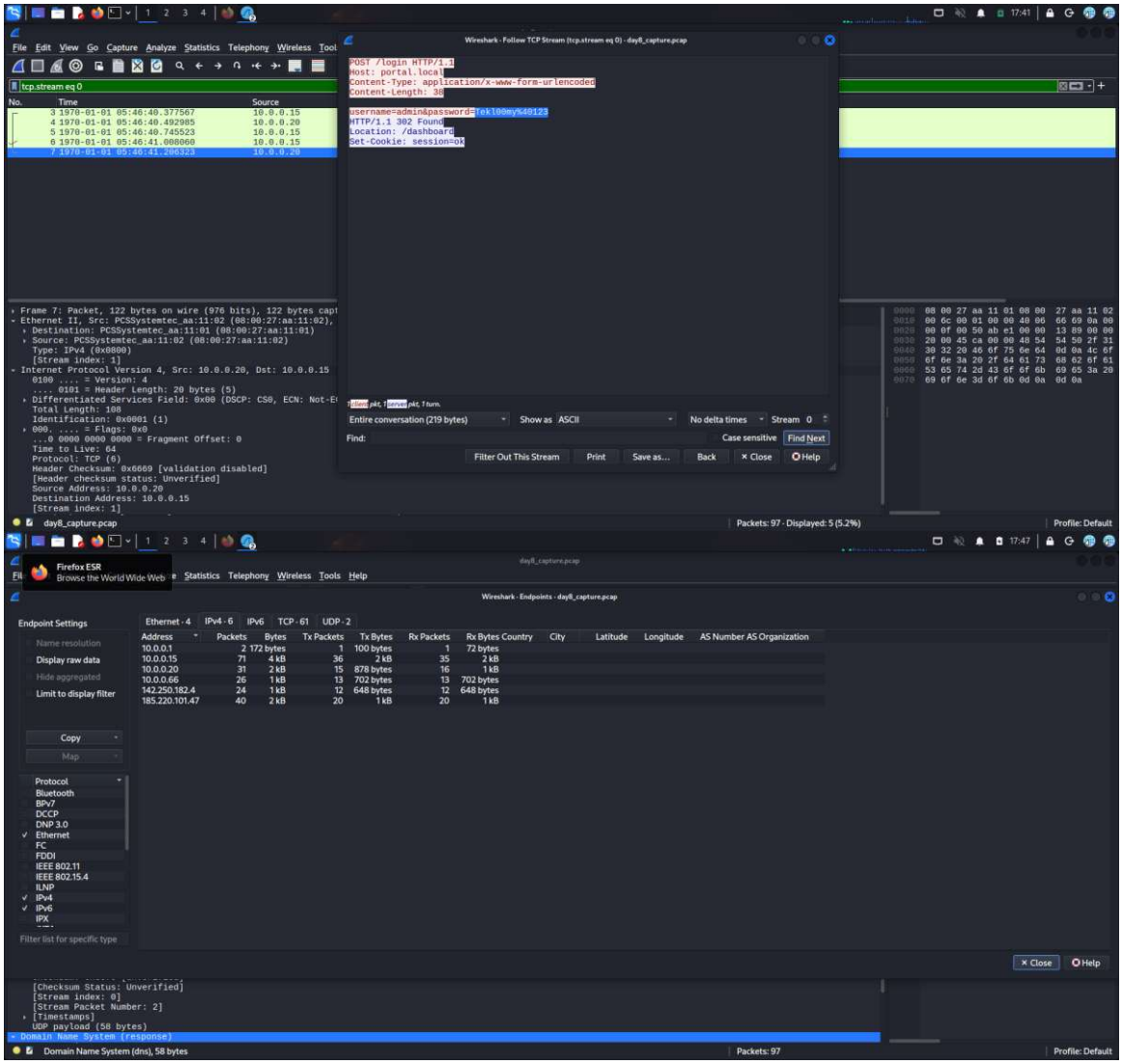
Question	Your answer
Which IP is doing the scanning (the attacker)?	10.0.0.15
Which IP is being scanned?	10.0.0.20
Roughly how many ports were probed?	26
Which ports replied SYN-ACK (open)?	Ports 22, 80, 443
What are those open ports normally used for?	22 is SSH (remote access), 80 is HTTP (web), 443 is HTTPS (secure web).

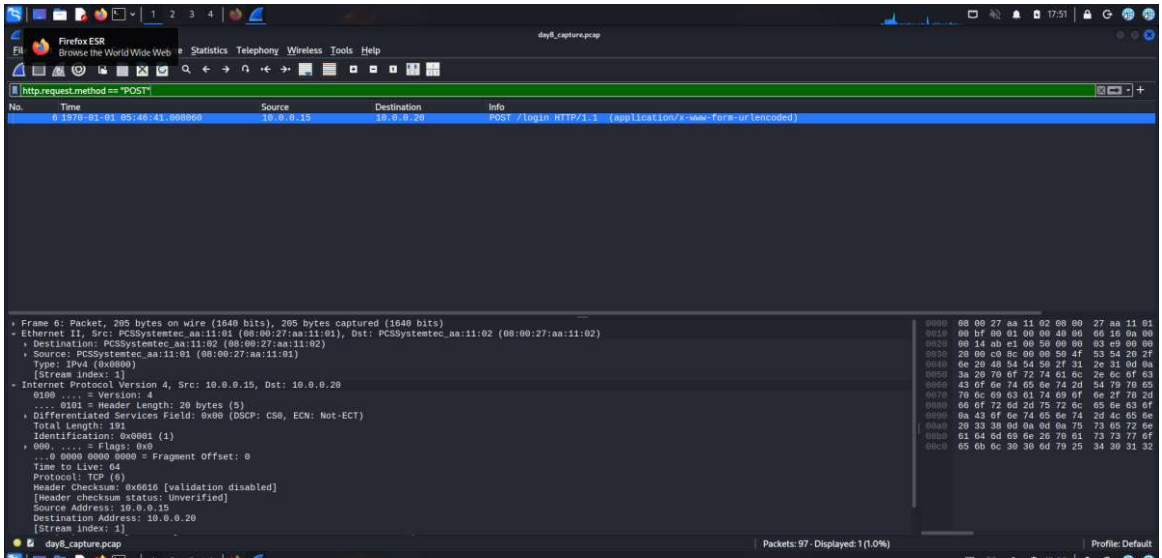
Finding 3 - The malware beacon (C2)

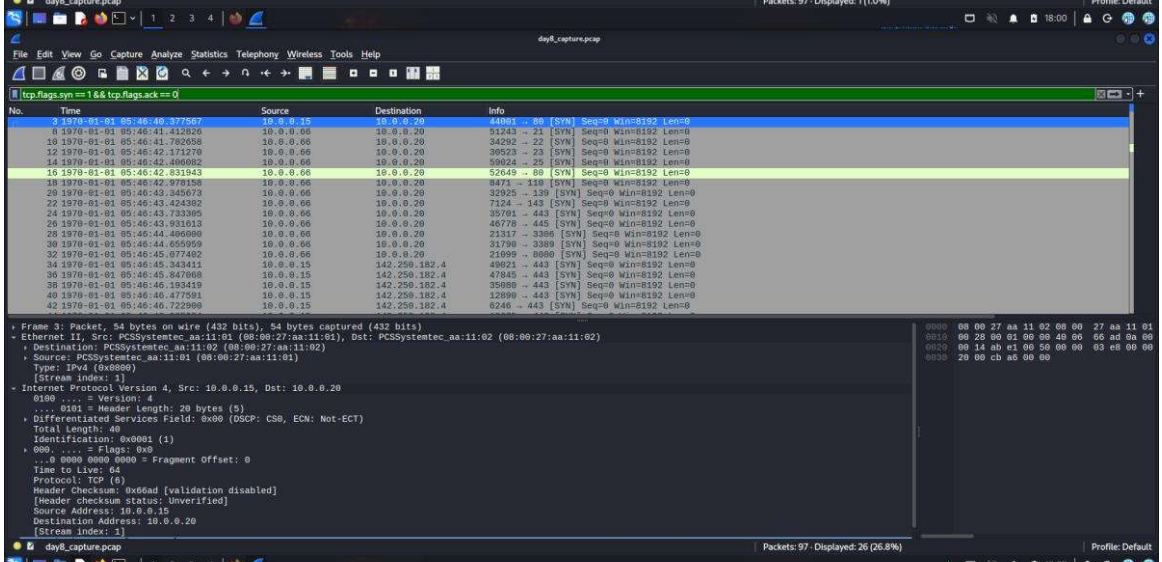
Question	Your answer
Which internal host is beaconing?	10.0.0.15
Which external IP is it contacting (the C2 server)?	142.250.182.4
On which destination port?	Port 443

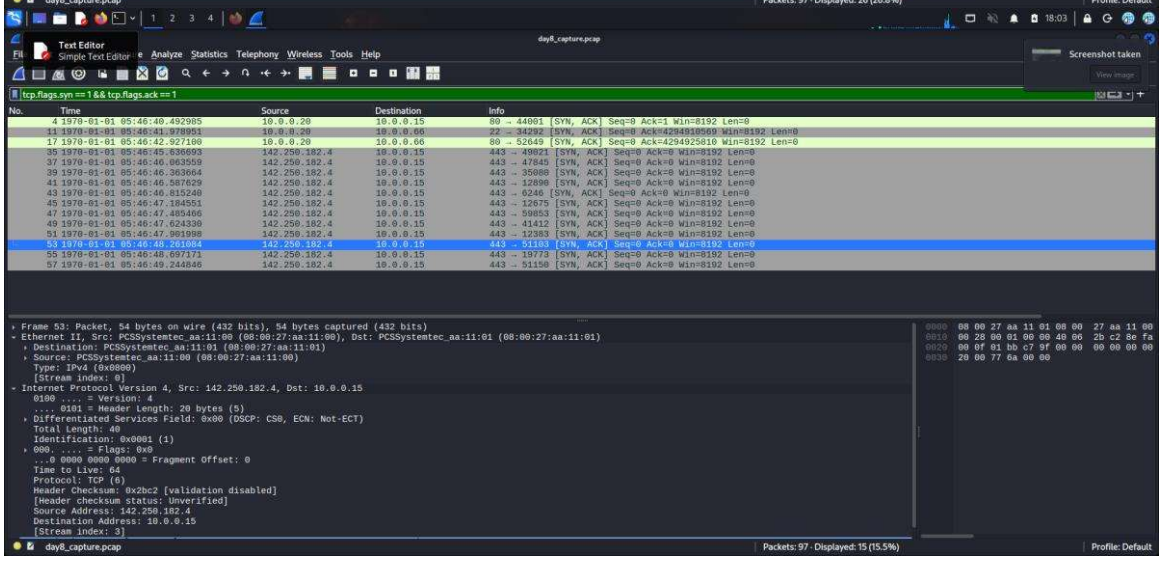
Roughly how often does it beacon (seconds between connections)?	Every 1 second
How is this different from the normal HTTPS traffic in the capture?	Normal HTTPS traffic involves variable packet sizes, diverse destinations, and random intervals driven by a human. This traffic is mathematically regular to a single IP.

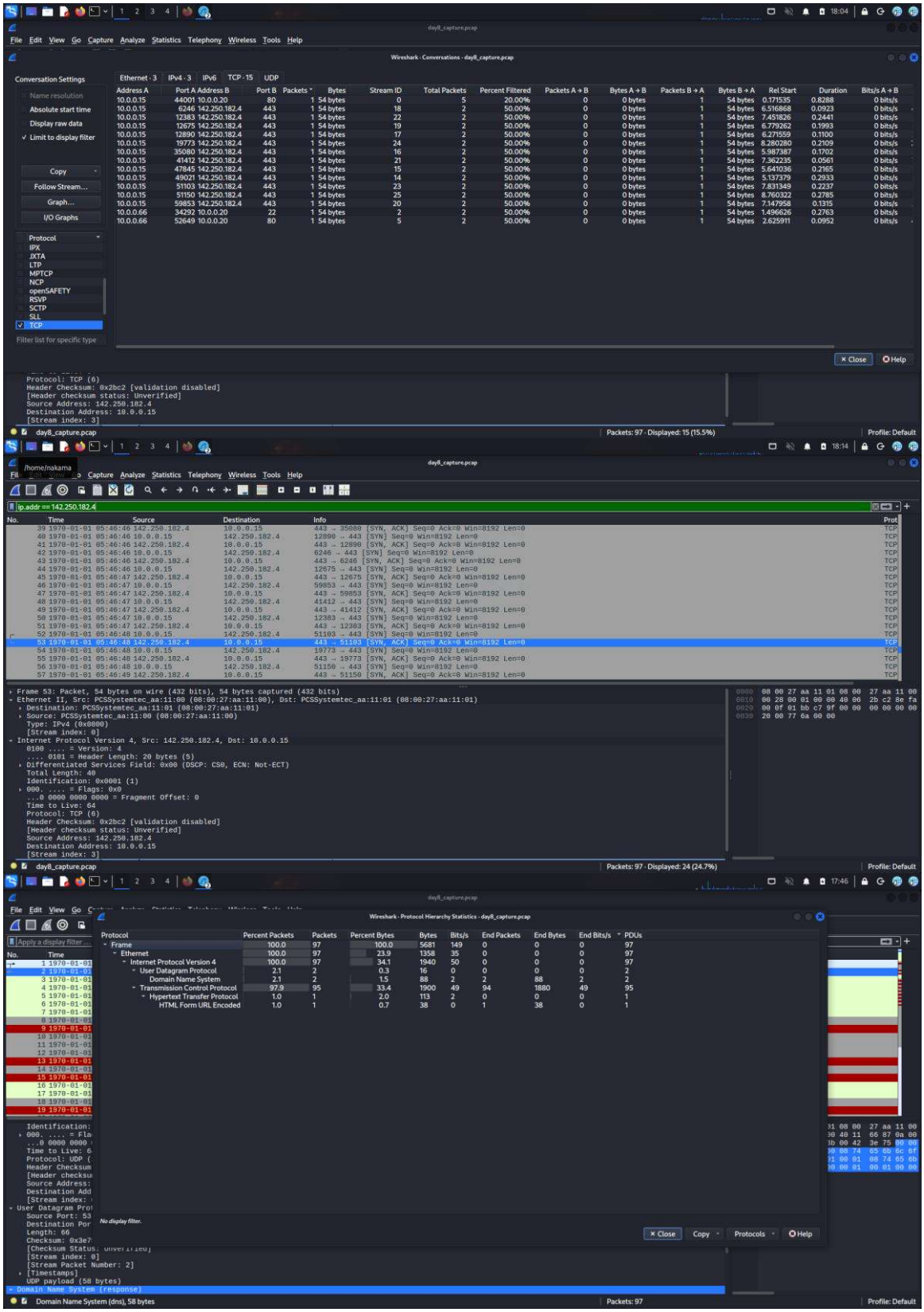
These are the attached images for the action I performed:-




Firefox ESR
Browse the World Wide Web
day8_capture.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method == "POST"
No. Time Source Destination Info
6 1970-01-01 05:46:41.008060 10.0.0.15 10.0.0.20 POST /login HTTP/1.1 (application/x-www-form-urlencoded)
Frame 6: Packet, 205 bytes on wire (1640 bits), 205 bytes captured (1640 bits)
Ethernet II, Src: PCSSystemtec_aa:11:01 (08:00:27:aa:11:01), Dst: PCSSystemtec_aa:11:02 (08:00:27:aa:11:02)
Destination: PCSSystemtec_aa:11:02 (08:00:27:aa:11:02)
Source: PCSSystemtec_aa:11:01 (08:00:27:aa:11:01)
Type: IPv4 (0x0800)
[Stream index: 1]
Internet Protocol Version 4, Src: 10.0.0.15, Dst: 10.0.0.20
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 191
Identification: 0x0001 (1)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x6616 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.0.15
Destination Address: 10.0.0.20
[Stream index: 1]
0000 08 00 27 aa 11 02 08 00 27 aa 11 01
0010 00 bf 00 01 00 00 40 06 66 16 0a 00
0020 00 14 ab e1 00 50 00 00 03 e9 00 00
0030 20 00 c0 8c 00 00 50 4f 53 54 20 2f
0040 6e 20 48 54 54 50 2f 31 2e 31 0d 0a
0050 3a 20 70 6f 72 74 61 6c 2e 6c 6f 63
0060 43 6f 6e 74 65 6e 74 2d 54 79 70 65
0070 70 6c 69 63 61 74 69 6f 6e 2f 70 2d
0080 66 6f 72 6d 2d 75 72 6c 65 6e 63 6f
0090 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e
00a0 20 33 30 0d 0a 0d 0a 75 73 65 72 6e
00b0 61 64 6d 69 6e 26 70 61 73 73 77 6f
00c0 65 6b 6c 30 30 6d 79 25 34 30 31 32
day8_capture.pcap
Packets: 97 · Displayed: 1 (1.0%)
Profile: Default


day8_capture.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.flags.syn == 1 && tcp.flags.ack == 0
No. Time Source Destination Info
3 1970-01-01 05:46:40.377567 10.0.0.15 10.0.0.20 44001 → 80 [SYN] Seq=0 Win=8192 Len=0
8 1970-01-01 05:46:41.412826 10.0.0.66 10.0.0.20 51243 → 21 [SYN] Seq=0 Win=8192 Len=0
10 1970-01-01 05:46:41.782658 10.0.0.66 10.0.0.20 34292 → 22 [SYN] Seq=0 Win=8192 Len=0
12 1970-01-01 05:46:42.171270 10.0.0.66 10.0.0.20 30523 → 23 [SYN] Seq=0 Win=8192 Len=0
14 1970-01-01 05:46:42.496082 10.0.0.66 10.0.0.20 59024 → 25 [SYN] Seq=0 Win=8192 Len=0
16 1970-01-01 05:46:42.831943 10.0.0.66 10.0.0.20 52649 → 80 [SYN] Seq=0 Win=8192 Len=0
18 1970-01-01 05:46:42.978158 10.0.0.66 10.0.0.20 8471 → 110 [SYN] Seq=0 Win=8192 Len=0
20 1970-01-01 05:46:43.345673 10.0.0.66 10.0.0.20 32925 → 139 [SYN] Seq=0 Win=8192 Len=0
22 1970-01-01 05:46:43.424302 10.0.0.66 10.0.0.20 7124 → 143 [SYN] Seq=0 Win=8192 Len=0
24 1970-01-01 05:46:43.733305 10.0.0.66 10.0.0.20 35701 → 443 [SYN] Seq=0 Win=8192 Len=0
26 1970-01-01 05:46:43.931613 10.0.0.66 10.0.0.20 46778 → 445 [SYN] Seq=0 Win=8192 Len=0
28 1970-01-01 05:46:44.406000 10.0.0.66 10.0.0.20 21317 → 3306 [SYN] Seq=0 Win=8192 Len=0
30 1970-01-01 05:46:44.655959 10.0.0.66 10.0.0.20 31790 → 3389 [SYN] Seq=0 Win=8192 Len=0
32 1970-01-01 05:46:45.077402 10.0.0.66 10.0.0.20 21099 → 8080 [SYN] Seq=0 Win=8192 Len=0
34 1970-01-01 05:46:45.343411 10.0.0.15 142.250.182.4 49021 → 443 [SYN] Seq=0 Win=8192 Len=0
36 1970-01-01 05:46:45.847068 10.0.0.15 142.250.182.4 47845 → 443 [SYN] Seq=0 Win=8192 Len=0
38 1970-01-01 05:46:46.193419 10.0.0.15 142.250.182.4 35080 → 443 [SYN] Seq=0 Win=8192 Len=0
40 1970-01-01 05:46:46.477591 10.0.0.15 142.250.182.4 12890 → 443 [SYN] Seq=0 Win=8192 Len=0
42 1970-01-01 05:46:46.722900 10.0.0.15 142.250.182.4 6246 → 443 [SYN] Seq=0 Win=8192 Len=0
Frame 3: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits)
Ethernet II, Src: PCSSystemtec_aa:11:01 (08:00:27:aa:11:01), Dst: PCSSystemtec_aa:11:02 (08:00:27:aa:11:02)
Destination: PCSSystemtec_aa:11:02 (08:00:27:aa:11:02)
Source: PCSSystemtec_aa:11:01 (08:00:27:aa:11:01)
Type: IPv4 (0x0800)
[Stream index: 1]
Internet Protocol Version 4, Src: 10.0.0.15, Dst: 10.0.0.20
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 40
Identification: 0x0001 (1)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x66ad [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.0.15
Destination Address: 10.0.0.20
[Stream index: 1]
0000 08 00 27 aa 11 02 08 00 27 aa 11 01
0010 00 28 00 01 00 00 40 06 66 ad 0a 00
0020 00 14 ab e1 00 50 00 00 03 e8 00 00
0030 20 00 cb a6 00 00
day8_capture.pcap
Packets: 97 · Displayed: 26 (26.8%)
Profile: Default


Text Editor
Simple Text Editor
day8_capture.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Screenshot taken
View image
tcp.flags.syn == 1 && tcp.flags.ack == 1
No. Time Source Destination Info
4 1970-01-01 05:46:40.492985 10.0.0.20 10.0.0.15 80 → 44001 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0
11 1970-01-01 05:46:41.978951 10.0.0.20 10.0.0.66 22 → 34292 [SYN, ACK] Seq=0 Ack=4294916569 Win=8192 Len=0
17 1970-01-01 05:46:42.927100 10.0.0.20 10.0.0.66 80 → 52649 [SYN, ACK] Seq=0 Ack=4294925810 Win=8192 Len=0
35 1970-01-01 05:46:45.636693 142.250.182.4 10.0.0.15 443 → 49021 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
37 1970-01-01 05:46:46.063550 142.250.182.4 10.0.0.15 443 → 47845 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
39 1970-01-01 05:46:46.363664 142.250.182.4 10.0.0.15 443 → 35080 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
41 1970-01-01 05:46:46.587629 142.250.182.4 10.0.0.15 443 → 12890 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
43 1970-01-01 05:46:46.815240 142.250.182.4 10.0.0.15 443 → 6246 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
45 1970-01-01 05:46:47.184551 142.250.182.4 10.0.0.15 443 → 12675 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
47 1970-01-01 05:46:47.485466 142.250.182.4 10.0.0.15 443 → 59853 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
49 1970-01-01 05:46:47.624330 142.250.182.4 10.0.0.15 443 → 41412 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
51 1970-01-01 05:46:47.901908 142.250.182.4 10.0.0.15 443 → 12383 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
53 1970-01-01 05:46:48.261004 142.250.182.4 10.0.0.15 443 → 51103 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
55 1970-01-01 05:46:48.697171 142.250.182.4 10.0.0.15 443 → 19773 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
57 1970-01-01 05:46:49.244846 142.250.182.4 10.0.0.15 443 → 51150 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0
Frame 53: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits)
Ethernet II, Src: PCSSystemtec_aa:11:00 (08:00:27:aa:11:00), Dst: PCSSystemtec_aa:11:01 (08:00:27:aa:11:01)
Destination: PCSSystemtec_aa:11:01 (08:00:27:aa:11:01)
Source: PCSSystemtec_aa:11:00 (08:00:27:aa:11:00)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 142.250.182.4, Dst: 10.0.0.15
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 40
Identification: 0x0001 (1)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x2bc2 [validation disabled]
[Header checksum status: Unverified]
Source Address: 142.250.182.4
Destination Address: 10.0.0.15
[Stream index: 3]
0000 08 00 27 aa 11 01 08 00 27 aa 11 00
0010 00 28 00 01 00 00 40 06 2b c2 8e fa
0020 00 0f 01 bb c7 9f 00 00 00 00 00 00
0030 20 00 77 6a 00 00
day8_capture.pcap
Packets: 97 · Displayed: 15 (15.5%)
Profile: Default


day8_capture.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Wireshark · Conversations · day8_capture.pcap
Conversation Settings
Ethernet · 3
IPv4 · 3
IPv6
TCP · 15
UDP
Address A
Port A
Address B
Port B
Packets
Bytes
Stream ID
Total Packets
Percent Filtered
Packets A → B
Bytes A → B
Packets B → A
Bytes B → A
Rel Start
Duration
Bits/s A → B
10.0.0.15 44001 10.0.0.20 80 1 54 bytes 0 5 20.00% 0 0 bytes 1 54 bytes 0.171535 0.8288 0 bits/s
10.0.0.15 6246 142.250.182.4 443 1 54 bytes 18 2 50.00% 0 0 bytes 1 54 bytes 6.516868 0.0923 0 bits/s
10.0.0.15 12383 142.250.182.4 443 1 54 bytes 22 2 50.00% 0 0 bytes 1 54 bytes 7.451826 0.2441 0 bits/s
10.0.0.15 12675 142.250.182.4 443 1 54 bytes 19 2 50.00% 0 0 bytes 1 54 bytes 6.779262 0.1993 0 bits/s
10.0.0.15 12890 142.250.182.4 443 1 54 bytes 17 2 50.00% 0 0 bytes 1 54 bytes 6.271559 0.1100 0 bits/s
10.0.0.15 19773 142.250.182.4 443 1 54 bytes 24 2 50.00% 0 0 bytes 1 54 bytes 8.280280 0.2109 0 bits/s
10.0.0.15 35080 142.250.182.4 443 1 54 bytes 16 2 50.00% 0 0 bytes 1 54 bytes 5.987387 0.1702 0 bits/s
10.0.0.15 41412 142.250.182.4 443 1 54 bytes 21 2 50.00% 0 0 bytes 1 54 bytes 7.362235 0.0561 0 bits/s
10.0.0.15 47845 142.250.182.4 443 1 54 bytes 15 2 50.00% 0 0 bytes 1 54 bytes 5.641036 0.2165 0 bits/s
10.0.0.15 49021 142.250.182.4 443 1 54 bytes 14 2 50.00% 0 0 bytes 1 54 bytes 5.137379 0.2933 0 bits/s
10.0.0.15 51103 142.250.182.4 443 1 54 bytes 23 2 50.00% 0 0 bytes 1 54 bytes 7.831349 0.2237 0 bits/s
10.0.0.15 51150 142.250.182.4 443 1 54 bytes 25 2 50.00% 0 0 bytes 1 54 bytes 8.760322 0.2785 0 bits/s
10.0.0.15 59853 142.250.182.4 443 1 54 bytes 20 2 50.00% 0 0 bytes 1 54 bytes 7.147958 0.1315 0 bits/s
10.0.0.66 34292 10.0.0.20 22 1 54 bytes 2 2 50.00% 0 0 bytes 1 54 bytes 1.496626 0.2763 0 bits/s
10.0.0.66 52649 10.0.0.20 80 1 54 bytes 5 2 50.00% 0 0 bytes 1 54 bytes 2.625911 0.0952 0 bits/s
Name resolution
Absolute start time
Display raw data
Limit to display filter
Copy
Follow Stream...
Graph...
I/O Graphs
Protocol
IPX
JXTA
LTP
MPTCP
NCP
openSAFETY
RSVP
SCTP
SLL
TCP
Filter list for specific type
Close
Help
Protocol: TCP (6)
Header Checksum: 0x2bc2 [validation disabled]
[Header checksum status: Unverified]
Source Address: 142.250.182.4
Destination Address: 10.0.0.15
[Stream index: 3]
day8_capture.pcap
Packets: 97 · Displayed: 15 (15.5%)
Profile: Default
/home/nakama
ip.addr == 142.250.182.4
No. Time Source Destination Info Prot
39 1970-01-01 05:46:46 142.250.182.4 10.0.0.15 443 → 35080 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
40 1970-01-01 05:46:46 10.0.0.15 142.250.182.4 12890 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
41 1970-01-01 05:46:46 142.250.182.4 10.0.0.15 443 → 12890 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
42 1970-01-01 05:46:46 10.0.0.15 142.250.182.4 6246 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
43 1970-01-01 05:46:46 142.250.182.4 10.0.0.15 443 → 6246 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
44 1970-01-01 05:46:46 10.0.0.15 142.250.182.4 12675 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
45 1970-01-01 05:46:47 142.250.182.4 10.0.0.15 443 → 12675 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
46 1970-01-01 05:46:47 10.0.0.15 142.250.182.4 59853 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
47 1970-01-01 05:46:47 142.250.182.4 10.0.0.15 443 → 59853 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
48 1970-01-01 05:46:47 10.0.0.15 142.250.182.4 41412 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
49 1970-01-01 05:46:47 142.250.182.4 10.0.0.15 443 → 41412 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
50 1970-01-01 05:46:47 10.0.0.15 142.250.182.4 12383 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
51 1970-01-01 05:46:47 142.250.182.4 10.0.0.15 443 → 12383 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
52 1970-01-01 05:46:48 10.0.0.15 142.250.182.4 51103 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
53 1970-01-01 05:46:48 142.250.182.4 10.0.0.15 443 → 51103 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
54 1970-01-01 05:46:48 10.0.0.15 142.250.182.4 19773 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
55 1970-01-01 05:46:48 142.250.182.4 10.0.0.15 443 → 19773 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
56 1970-01-01 05:46:48 10.0.0.15 142.250.182.4 51150 → 443 [SYN] Seq=0 Win=8192 Len=0 TCP
57 1970-01-01 05:46:49 142.250.182.4 10.0.0.15 443 → 51150 [SYN, ACK] Seq=0 Ack=0 Win=8192 Len=0 TCP
Frame 53: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits)
Ethernet II, Src: PCSSystemtec_aa:11:00 (08:00:27:aa:11:00), Dst: PCSSystemtec_aa:11:01 (08:00:27:aa:11:01)
Destination: PCSSystemtec_aa:11:01 (08:00:27:aa:11:01)
Source: PCSSystemtec_aa:11:00 (08:00:27:aa:11:00)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 142.250.182.4, Dst: 10.0.0.15
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 40
Identification: 0x0001 (1)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x2bc2 [validation disabled]
[Header checksum status: Unverified]
Source Address: 142.250.182.4
Destination Address: 10.0.0.15
[Stream index: 3]
0000 08 00 27 aa 11 01 08 00 27 aa 11 00
0010 00 28 00 01 00 00 40 06 2b c2 8e fa
0020 00 0f 01 bb c7 9f 00 00 00 00 00 00
0030 20 00 77 6a 00 00
day8_capture.pcap
Packets: 97 · Displayed: 24 (24.7%)
Profile: Default
Wireshark · Protocol Hierarchy Statistics · day8_capture.pcap
Protocol
Percent Packets
Packets
Percent Bytes
Bytes
Bits/s
End Packets
End Bytes
End Bits/s
PDUs
Frame 100.0 97 100.0 5681 149 0 0 0 97
Ethernet 100.0 97 23.9 1358 35 0 0 0 97
Internet Protocol Version 4 100.0 97 34.1 1940 50 0 0 0 97
User Datagram Protocol 2.1 2 0.3 16 0 0 0 0 2
Domain Name System 2.1 2 1.5 88 2 2 88 2 2
Transmission Control Protocol 97.9 95 33.4 1900 49 94 1880 49 95
Hypertext Transfer Protocol 1.0 1 2.0 113 2 0 0 0 1
HTML Form URL Encoded 1.0 1 0.7 38 0 1 38 0 1
No display filter.
Close
Copy
Protocols
Help
Domain Name System (response)
Domain Name System (dns), 58 bytes
Packets: 97
Profile: Default

Step 4 Write the mini incident report

To the management team: During a network review, I identified that the internal workstation at 10.0.0.15 has been compromised. First, the user logged into our internal web server over an unencrypted HTTP connection, leaking their admin credentials. Second, the compromised machine (10.0.0.15) performed a port scan against our server (10.0.0.20). Finally, it established a steady automated connection (C2 beacon) to an external malicious server on port 443. My immediate actions will be to:

1) Isolate 10.0.0.15 from the network,

2) Force a password reset for the compromised admin account

3) Block the external C2 IP address at our firewall.